MILITARY SPECIFICATION

COLLIMATOR, VISUAL SU-102/UA

This specification is approved for use by USAERADCOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of common module, Collimator, Visual SU-102/UA.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of
MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment
MIL-E-55585	- Electronic Equipment and Parts, Packaging of

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-150	- Photographic Lenses
MIL-STD-252	- Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices
MIL-STD-454	- Standard General Requirements for Electronic Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, USA Electronics Research and Development Command, ATTN: DRDEL-E, 2800 Powder Mill Road, Adelphi, MD 20783 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5855

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MIL-STD-726	- Packaging Requirement Codes
MIL-STD-810	- Environmental Test Methods

DRAWINGS

USA ELECTRONICS COMMAND

DLSM-B-773203	- Collimator, Visual SU-102/UA
SM-D-806557-6	- Field Flattener, Emitter Module

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The Collimator, Visual SU-102/UA, referred to herein as the collimator, consists of optical elements which collimate the visual imagery within an infrared system.

3.2 Construction. The collimator shall be constructed in accordance with DLSM-B-773203 and as specified herein.

3.2.1 Weight. The weight of the collimator shall be not greater than 1.07 pounds.

3.3 First article. When specified in the contract (see 6.2.b) the contractor shall furnish first article units for inspection (see 4.3).

3.4 Materials, parts, and processes. Materials, parts, and processes shall be as specified herein and as shown on the applicable drawings. Materials, parts, and processes not specified shall be selected by the contractor in accordance with MIL-P-11268.

3.5 Components. The collimator shall consist of optical elements, a flat mirror, and a housing.

3.6 Performance characteristics.

3.6.1 Effective focal length. The effective focal length of the collimator in the 630 to 690 nanometer spectral region shall be 67.8, ±0.7 millimeters.

3.6.2 Effective aperture. The effective aperture of the collimator shall not be less than 39.4 millimeters.

3.6.3 Linear distortion. The linear distortion of the collimator shall be 0 to -2 percent at the edge of the field (20 millimeters diameter).

3.6.4 Modulation transfer function (MTF). The MTF of the collimator shall not be less than the MTF given in figures 1 and 2, measured in a flat-field focal plane and parallel to surface B in figure 3.

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3.6.5 Stray light veiling glare. The stray light veiling glare of the collimator shall not be greater than 5 percent.

3.6.6 Relative illumination. The relative illumination of the collimator shall not be less than 60 percent at the edge of the field, for a field diameter of 20 millimeters at the image plane.

3.6.7 Flange focal distance.

3.6.7.1 Flange focal distance setting. The flange focal distance of the collimator shall be 11.91, ±0.25 millimeters. The flange surface is shown as surface B in figure 3.

3.6.7.2 Flange focal distance variation. The specific value of the collimator flange focal distance shall not vary more than ±0.05 millimeter over the high operating temperature (+23°C to +71°C) and shall remain within the required 11.91, ±0.25 millimeter setting.

3.6.8 Optical transmittance. The optical transmittance of the collimator, measured in the 630- to 690-nanometer spectral region and excluding the field flattener lens, shall not be less than 85 percent.

3.6.9 Deviation. The output optical axis of the collimator shall not deviate from the 90.0 degree input optical axis by more than ±0.5 degree.

3.7 Environmental conditions.

3.7.1 Temperature shock. The collimator shall not be damaged (see 6.3.1) by sudden changes in temperature between -54°C and +95°C.

3.7.2 High temperature. The collimator shall not be damaged by storage to +95°C or operation to +71°C.

3.7.3 Low temperature. The collimator shall not be damaged by operation to -54°C or storage to -62°C.

3.7.4 Shock. The collimator shall not be damaged by high intensity shocks of 100 g's peak amplitude with 11 milliseconds duration and by bench handling tests.

3.7.5 Vibration. The collimator shall not be damaged after sustaining vibration over a frequency spectrum at the specified g levels and amplitudes shown in figure 4.

3.7.6 Altitude. The collimator shall not be damaged by operation at altitudes up to 10,000 feet above sea level or storage to 50,000 feet above sea level.

3.8 Treatment and painting. Unless otherwise specified, the collimator and its components and parts shall be cleaned, treated and painted in accordance with the applicable drawing.

3.9 Nameplates and product marking. Unless otherwise specified, the collimator, parts, components, subassemblies, and assemblies thereof shall be marked for identification and reference designation markings in accordance with the applicable drawing.

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3.10 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.5).
c. Inspection of packaging (see 4.8).

4.3 First article inspection. When specified in the contract, first article inspection shall be performed by the contractor on a minimum of 3 sample units.

4.3.1 Inspections. All materials, parts, processes and assemblies shall be examined for conformance to the applicable specification or drawing. Inspections shall be made using MIL-STD-105, General Inspection Level II with a 2.5 percent AQL for mechanical and a 4 percent AQL for visual inspection, except as noted in table I.

TABLE I. Inspections.

Inspection Requirement	Percent AQL	Requirement paragraph	Inspection criteria
Any part or component missing or damaged	2.5	3.2, 3.4	MIL-STD-252
Weight not as specified	2.5	3.2.1	MIL-STD-252
Marking not as specified	2.5	3.9	
Workmanship not as specified	2.5	3.10	MIL-STD-454
Treatment and painting not as specified	2.5 1/	3.8	
Dimensions not as specified		3.2	SM-D-773417
Flange diameter (2.000 in. Ref)	1.0		SM-D-773398
Housing diameter, scanner side (1.999 in. Ref)	1.0		

1/ MIL-STD-105, Inspection Level S-3 or objective quality evidence.

4.3.2 Tests. Upon successful completion of inspections specified in 4.3.1, perform the tests in table II in any order. Failure of any tests shall constitute first article failure.

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TABLE II. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Modulation transfer function	3.6.4	4.6.4.1
Flange focal distance	3.6.7	4.6.7
Effective focal length	3.6.1	4.6.1
Linear distortion	3.6.3	4.6.3
Effective aperture	3.6.2	4.6.2
Deviation	3.6.9	4.6.9
Optical transmittance	3.6.8	4.6.8
Relative illumination	3.6.6	4.6.6
Stray light veiling glare	3.6.5	4.6.5
High temperature	3.7.2	4.7.2
Temperature shock	3.7.1	4.7.1
Low temperature	3.7.3	4.7.3
Vibration	3.7.5	4.7.5
Shock	3.7.4	4.7.4
Altitude	3.7.6	4.7.6

4.3.3 Disposition of first article samples. First article samples shall not be considered as part of the procurement quantities (see 6.2.f).

4.4 Inspection procedures for quality assurance provisions. Unless otherwise specified herein, the collimator shall be tested at an ambient temperature of +23°C, ±5°C. Tests shall be conducted with radiation in the 630- to 690-nanometer spectral region and using a field flattener lens and clear aperture shown in figure 3. The test collimator used shall have a wavefront error of less than 0.25 wavelength (peak-to-peak) at 632.8 nanometers (Ref) over the clear circular aperture of the unit under test. Microscope objectives used to relay imagery shall have a numerical aperture (N.A.) of not less than 0.3. The test light source condenser N.A. at the test object shall not be less than twice the projected N.A. of the unit under test. Linearity of photometric measurement setups shall be within 5 percent.

4.5 Quality conformance inspection.

4.5.1 Inspection. Inspection shall be as specified in 4.3.1.

4.5.2 Tests.

4.5.2.1 Group A inspection. Group A inspection shall be conducted at an ambient temperature of +23°C, ±5°C on all collimators following the inspection of 4.5.1. Group A tests listed in table III may be performed in any order. Failure of any test shall be cause for rejection of that unit.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Modulation transfer function	3.6.4	4.6.4.2
Flange focal distance	3.6.7	4.6.7

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4.5.2.2 Group B inspection. Not required.

4.5.2.3 Group C inspection. Group C inspection shall be conducted on collimators selected from units which have passed the tests in 4.5.2.1. The sample(s) shall be tested in accordance with inspections listed in table IV. Samples shall be selected in accordance with 4.5.2.3.1. Group C tests listed in table IV may be performed in any order.

4.5.2.3.1 Sampling for group C inspection. Group C inspection samples shall be selected as follows:

Monthly lot size	Number of sample units monthly
1-99	1
100-299	3
300-499	5
500-999	7
1000 or more	10

TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Straylight veiling glare	3.6.5	4.6.5
Relative illumination	3.6.6	4.6.6
Optical transmittance	3.6.8	4.6.8
Deviation	3.6.9	4.6.9
Temperature shock	3.7.1	4.7.1
Vibration	3.7.5	4.7.5

4.5.2.3.2 Group C failures. Actions required relative to group C failures shall be as specified in the contract (see 6.2.c.(2)).

4.5.2.3.3 Disposition of group C samples. Group C samples shall be accepted on contract subsequent to the tests of table III.

4.5.2.4 Group D ispection. Not required.

4.6 Test methods.

4.6.1 Effective focal length. Effective focal length shall be measured in accordance with MIL-STD-150, Method 1 or 2. Failure to meet requirements of 3.6.1 shall constitute failure of this test.

4.6.2 Effective aperture. Effective aperture shall be measured in accordance with MIL-STD-150, Method 3 or 4. Failure to meet requirements of 3.6.2 shall constitute failure of this test.

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4.6.3 Linear distortion. Linear distortion shall be measured in accordance with MIL-STD-150, Method 27 or 28. Failure to meet requirements of 3.6.3 shall constitute failure of this test.

4.6.4 Modulation transfer function. The MTF shall be measured using a test setup calibrated to provide frequency and modulus. The MTF shall be measured either continuously or in 10 discrete steps from 0 to 20 cycles per millimeter (cy/mm). The analyzed area shall not be less than 0.25 mm. The MTF test shall be run at infinite conjugates.

4.6.4.1 MTF first article. On-axis MTF shall be measured in two perpendicular orientations in the same focal plane and shall not be less than the MTF in figure 1.

4.6.4.2 MTF group A. On-axis MTF shall be measured in a single orientation and shall not be less than the MTF in figure 1. The off-axis radial MTF shall be measured in the same plane used for the on-axis image parallel to the plane defined by surface B of figure 3. The off-axis MTF shall be measured at 2 points in the field 90 degrees apart. Off-axis MTF shall not be less than the MTF in figure 2.

4.6.4.3 Failure criteria. Failure to meet requirements of 3.6.4 shall constitute failure of these tests.

4.6.5 Stray light veiling glare. Stray light veiling glare shall be measured without a field flattener lens using a 1 degree, ±5 minutes black spot in the emitter focal plane of the collimator, surrounded by a 20-millimeter diameter Lambertian field. The Lambertian field luminance shall not be less than 100 times that of the black spot. Stray light shall be measured with a telephotometer at the entrance pupil of the collimator as defined by figure 3. Stray light veiling glare shall be computed as the ratio of the luminance of the black spot to the surrounding-field luminance. Failure to meet requirements of 3.6.5 shall constitute failure of this test.

4.6.6 Relative illumination. Relative illumination shall be measured in accordance with MIL-STD-150, Method 8. Relative illumination shall be measured in 2 mutually perpendicular planes with the first plane measured across the greatest distance of the oval shape of the mirror. Failure to meet requirements of 3.6.6 shall constitute failure of this test.

4.6.7. Flange focal distance. Flange focal distance shall be measured from surface B in figure 3 to the flat-field MTF plane (see 4.6.4). Failure to meet requirements of 3.6.7 shall constitute failure of this test.

4.6.8 Optical transmittance. Optical transmittance shall be measured in accordance with MIL-STD-150, Method 5 or 6 using a beam diameter of at least 22 millimeters. The field flattener lens shall be removed for this test. Failure to meet requirements of 3.6.8 shall constitute failure of this test.

4.6.9 Deviation. The deviation shall be measured by determining the angle between the plane of flange B of figure 3 and the plane of best focus. The plane of best focus shall be determined by measuring the flange focal distance of four image points of 5 mm field height and 90 degrees apart. Best MTF at 10 cy/mm shall be used to determine the exact location of these points with respect to

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flange B. The deviation is a vector having components in the x and y directions. The deviation for a single component is determined by the arctangent of the difference in focal positions for opposite sides divided by the distance between the points. The deviation is then the square root of the sum of the squares of the individual components.

4.7 Environmental tests. Unless otherwise specified, environmental tests shall be performed in accordance with MIL-STD-810. Unless otherwise specified, operating tests required before and after these environmental tests shall be those listed in table III.

4.7.1 Temperature shock. The collimator shall be tested in accordance with MIL-STD-810, Method 503, Procedure I except that the temperature of step 2 shall be -54°C and the temperature of step 1 and step 4 shall be +95°C. Failure to meet requirements of 3.7.1 shall constitute failure of this test.

4.7.2 High temperature. The collimator shall be tested in accordance with MIL-STD-810, Method 501, Procedure I except that the temperature of step 2 shall be +95°C. The temperature of steps 4 and 5 shall be +71°C. Failure to meet requirements of 3.7.2 and 3.6.7.2 during the operating test shall constitute failure of this test.

4.7.3 Low temperature. The collimator shall be tested in accordance with MIL-STD-810, Method 502, Procedure I except that the storage temperature of step 2 shall be -62°C which shall be maintained until stabilization is reached. The low operating temperature of step 4 shall be -54°C. Operating tests shall not be performed during low temperature test. Failure to meet requirements of 3.7.3 shall constitute failure of this test.

4.7.4 Shock. The collimator shall be tested in accordance with MIL-STD-810, Method 516.1, Procedure IV, Figure 516.1-1 (100 g's peak value at 11 milliseconds time duration) and Procedure V. Operating tests shall not be performed during shock test. Failure to meet requirements of 3.7.4 shall constitute failure of this test.

4.7.5 Vibration. The collimator shall be tested in accordance with MIL-STD-810, Method 514.1, Procedures I and VIII except that the curve shown in figure 4 of this specification shall be used in lieu of the MIL-STD-810 curve. The sinusoidal cycle time shall be 120 minutes per axis. The dwell time shall be 1/6 of the cycling time at each resonance. Operating tests shall not be performed during vibration test. Failure to meet requirements of 3.7.5 shall constitute failure of this test.

4.7.6 Altitude. The collimator shall be tested in accordance with MIL-STD-810, Method 500, Procedure I. Operating tests shall not be performed during altitude test. Failure to meet the requirements of 3.7.6 shall constitute failure of this test.

4.8 Inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine compliance with requirements of section 5.

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5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. 2M-1-B-ØØ-00-NV-X-XX-G-ED-A.

5.1.2 Level B. 2M-1-B-ØØ-00-NV-X-XX-G-ED-B.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. The collimator is intended for use in infrared systems.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. First article tests are required except for existing suppliers (see 6.3.2).

(1) Time frame for submission of first article test reports when first article tests are required.

(2) Time frame for approval of first article test reports when first article tests are required.

c. Production delivery schedule.

(1) Defined in terms of monthly lots.

(2) Actions required relative to group C failures (see 4.5.2.3.2).

d. Level A or level B preservation and packaging (see section 5).
e. MIL-STD-810B shall be used for environmental tests of 4.7.
f. Deliver all first article samples to Commander, USAERADCOM, ATTN: DELNV-CM, Fort Belvoir, VA 22060.
g. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.
h. Group C inspection monthly sample size to be established based on cumulative totals of concurrent contracts.

6.3 Definitions.

6.3.1 Damage. Breakage, loosening, shifting, evidence of corrosion, or failure of any finish, hardware, connection or component; and any degradation of collimator performance to values less than specified herein.

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6.3.2 Existing suppliers. Suppliers who are currently supplying equipment to this specification with a minimum average monthly rate of 10 units per month over the most current 12-month period. If a supplier cannot maintain the minimum monthly average, a first article test performed annually shall sustain the existing supplier's status when performed on 3 units manufactured by the supplier explicitly for this test. Upon implementation of this specification, suppliers who are then supplying equipment to specification B2-28A050105, Collimator Visual SU-102/UA with the minimum monthly delivery rate specified above shall be considered as existing suppliers.

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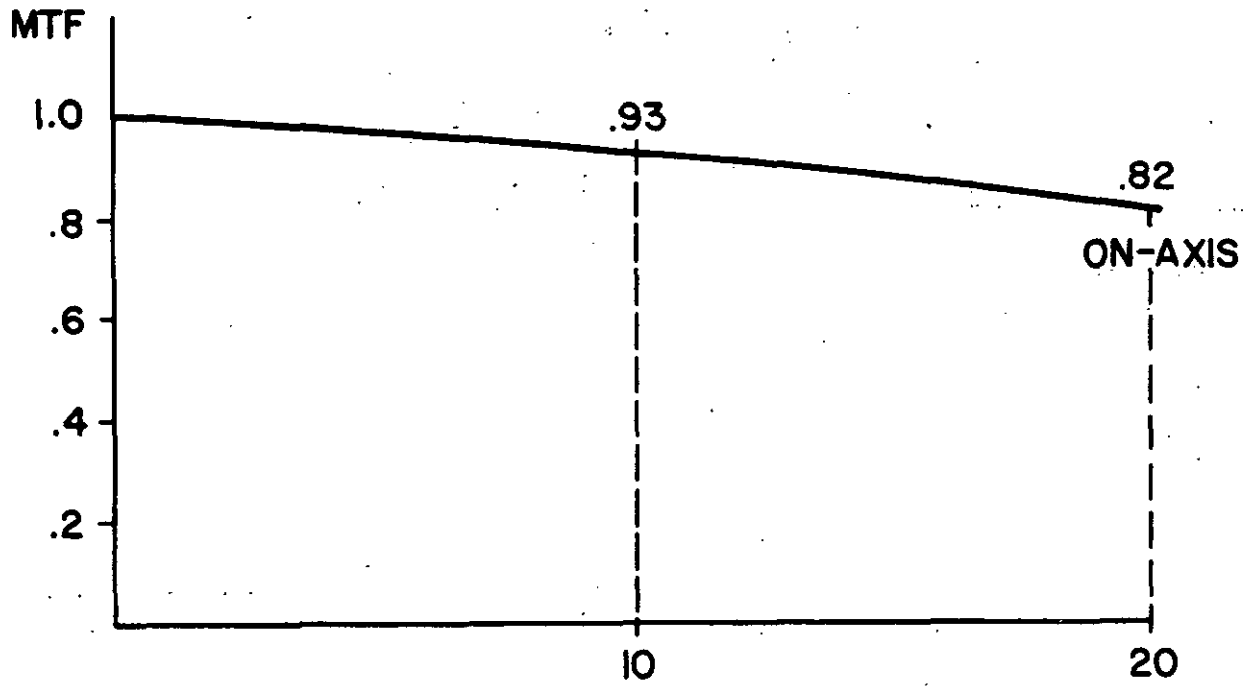


FIGURE 1. VISUAL COLLIMATOR ON-AXIS MTF.

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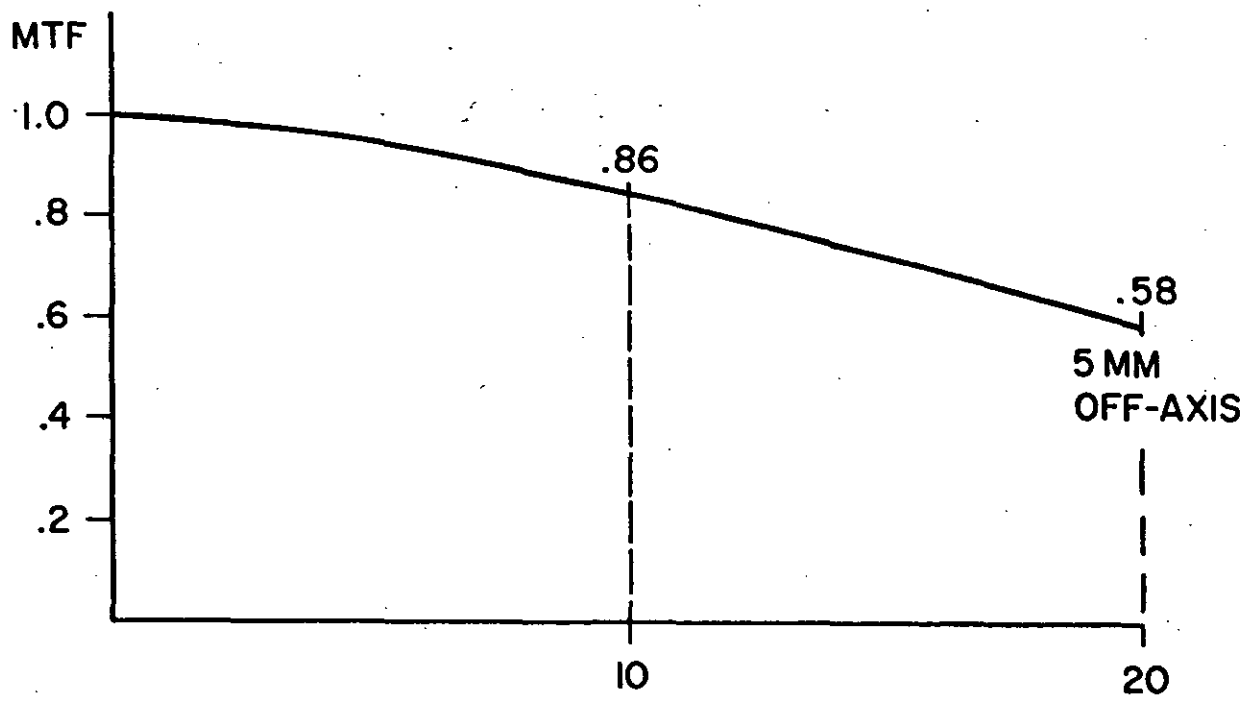


FIGURE 2. VISUAL COLLIMATOR OFF-AXIS MTF.

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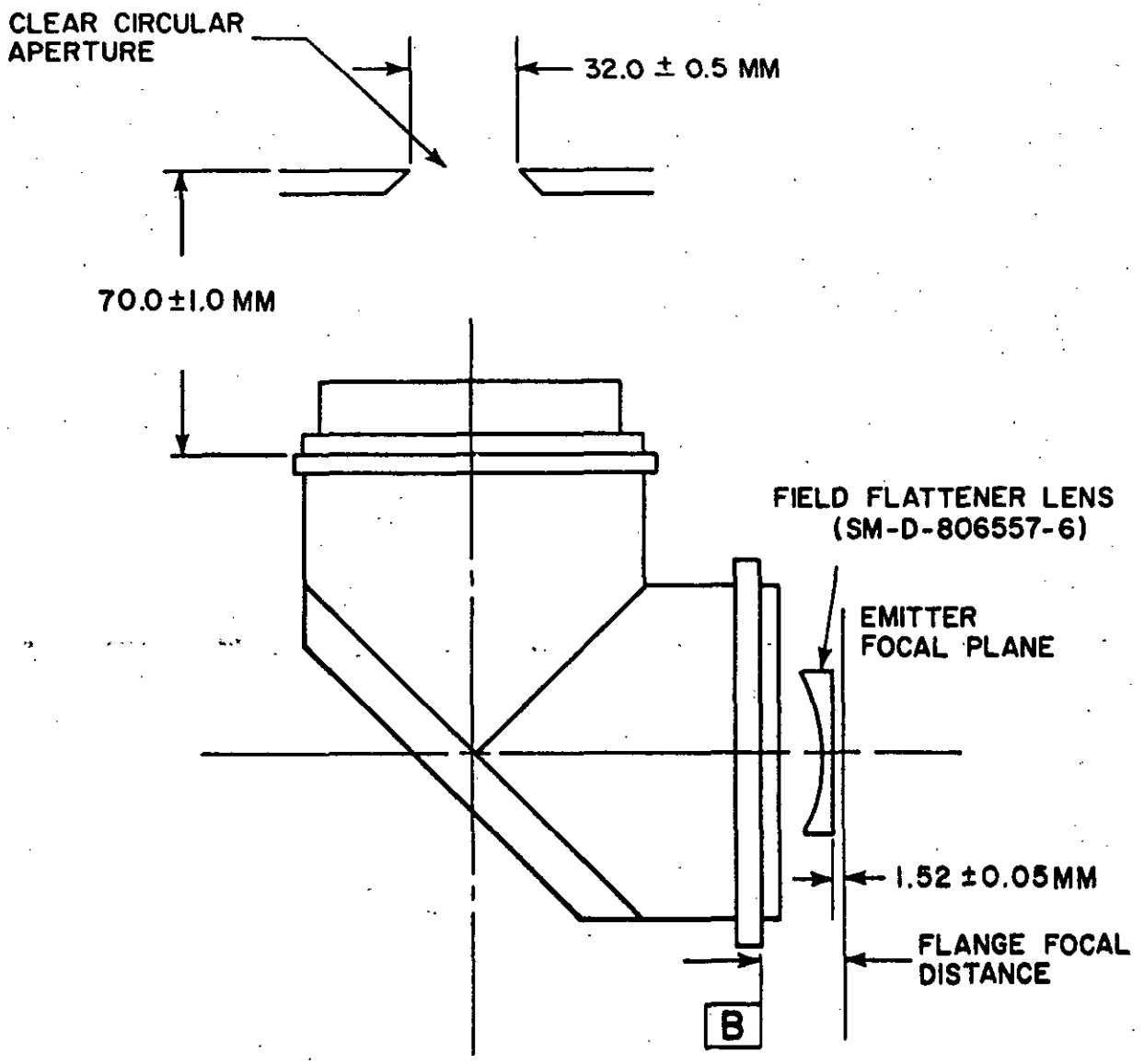


FIGURE 3. COLLIMATOR TEST CONFIGURATION.

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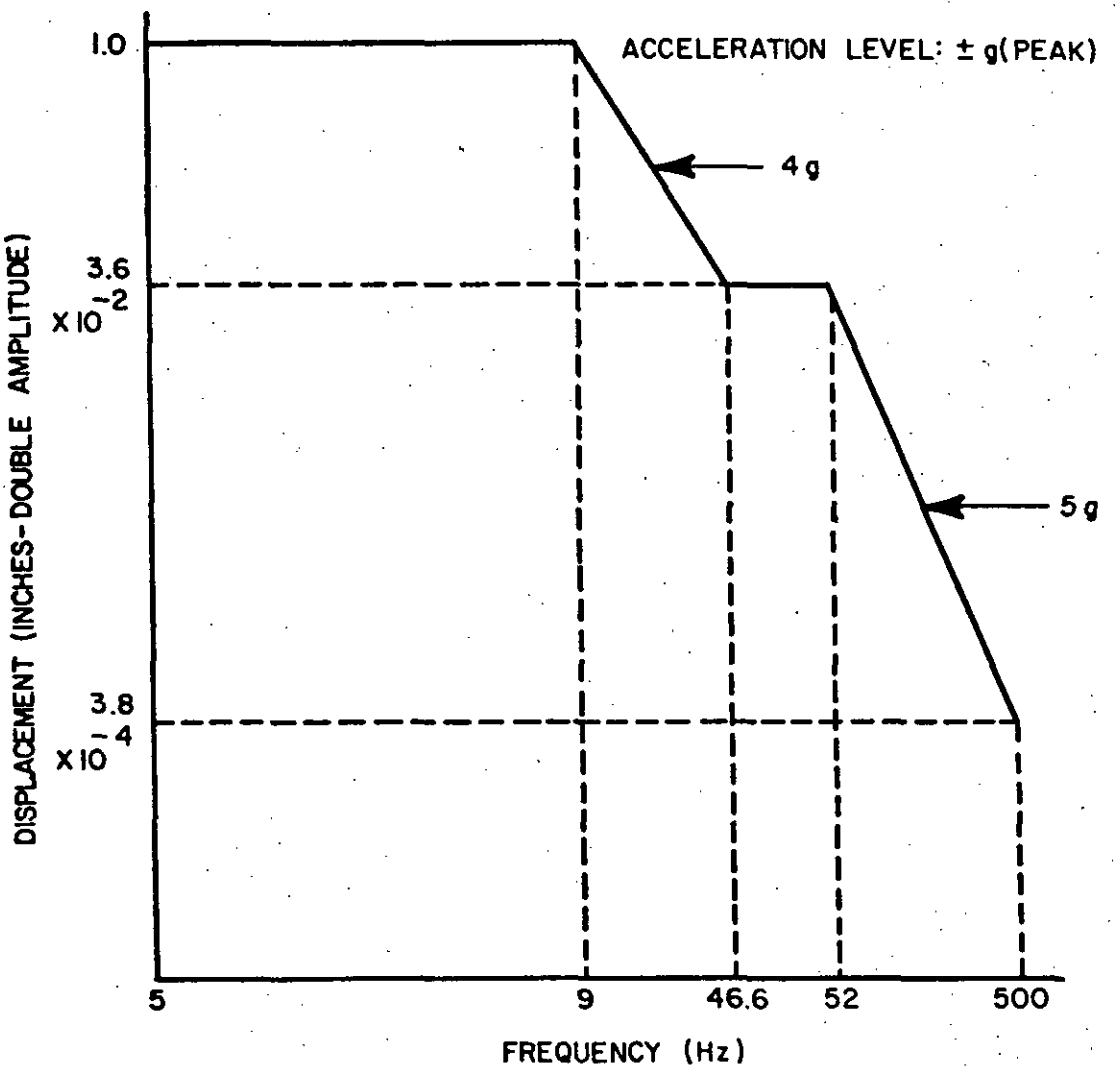


FIGURE 4. VIBRATION TEST PROFILE.

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DOCUMENT IDENTIFIER AND TITLE
MIL-C-49174A Collimator, Visual SU-102/UA

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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